QUANTITATIVE RESEARCH PORTFOLIO

Cross-Validated Pricing, Calibration and Hedging of SPX Options

Three independent pricing engines, a Heston / SABR / local-vol calibration to SPX, model risk on an autocallable, and a full-revaluation hedging attribution

ABSTRACT

A Fourier (COS), a Crank–Nicolson and a Monte Carlo engine are built and checked against each other and against published reference values on three test contracts. The validated library is then calibrated to end-of-day SPX option quotes. One Heston parameter set fits the 2023-01-18 surface to 0.89 implied-vol points, with its largest errors at one month; on the same surface κ is only weakly identified, and across 52 weekly fits in 2023 its changes move with those of ξ (correlation 0.96). For a 3-year autocallable, Heston and Dupire local vol imply fair coupons of 15.2% and 16.0%. A control model with Heston's own vanilla prices but local-vol dynamics attributes 0.77 of the 0.78-point gap to dynamics and 0.01 to calibration differences. In a daily-hedged SPX backtest, an exact full-revaluation split shows that the 2023 profit came from realised variance below implied ($+74.2$ points) partly offset by implied-vol re-marks (-34.2). It also shows that the large residual of the Greek (Taylor) attribution came mainly from the last three days before expiry.

CONTENTS

	Executive summary	1
1	Research design	2
2	Three engines, one price	3
3	Calibration to the SPX surface	5
4	Parameter identifiability and stability	7
5	Exotics and model risk	9
6	Greeks and the sources of hedging P&L	10
7–8	Evidence summary, limitations and next steps	13

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DATA

SPX end-of-day option quotes, optionsDX
Jan–Mar 2020 and Jan–Dec 2023
4.93 million quotes, 311 dates

CODE

github.com/kexinzhang2026/equity-derivatives-pricing

71 unit tests in continuous integration; one script per stage regenerates every table and figure.

Executive Summary

Research question. Do three independent numerical methods produce the same price, and once models are fitted to real SPX options, how much do the parts of each model that vanilla prices do not pin down move exotic prices and hedging P&L?

Table 1: Key results and the strength of the evidence behind each

Result	Evidence	Strength
Engines agree on the test contracts	BS call: 12.0056 closed form and COS, 12.0057 PDE, 12.0042 $\pm$ 0.0058 MC. Heston COS equals the published value to 2×10^{-8}; QE Monte Carlo within 0.2 s.e. PDE converges at order 2.00.	Strong, for the three contracts tested
Heston misses the short-dated SPX skew	RMSE 0.89 vol pt overall, 1.48 vol pt at one month; residuals structured by moneyness and maturity; same minimum from all four starts.	Strong for Heston; not a statement about all diffusions
κ is weakly identified	Real-surface profile: $\kappa \in [1.2, 3.0]$ fits within 0.1 vol pt of the optimum, about one half spread. Weekly changes of κ and ξ correlate at 0.96 (52 fits, 2023).	Strong on one date; supported over the year
Heston–local vol coupon gap is dynamics	Fair coupons 15.17% vs 15.95%. A local vol built from Heston's own vanillas gives 15.94%: dynamics $+0.77$, calibration $+0.01$ (MC s.e. about 0.19).	Supportive; one surface, one product
2023 short-vol P&L source	Full revaluation: realised vs implied variance $+74.2$, implied-vol re-marks -34.2, forward basis -8.2; total $+31.8$, exact.	Descriptive; 12 trades

WHAT THE EVIDENCE SUPPORTS

- Numerical results that can be trusted at the precision quoted: each engine is checked against a second engine or a published reference before it is used. Heston exotic prices rest on the validated Monte Carlo engine alone.
- A specific, testable account of Heston's SPX misfit: the error is concentrated at short maturities and is systematic in moneyness, the pattern expected from a one-factor diffusive stochastic-vol model.
- A controlled separation of model risk on the autocallable into calibration and dynamics, rather than attributing the whole gap to the forward smile by assumption.
- An exact P&L split for the hedging backtest, and an explanation of where the Taylor attribution fails.

LIMITS OF THE EVIDENCE

- No model fits the surface inside the bid–ask spread (median half spread 0.07–0.37 vol pt): Heston 0.83 vol pt, local vol 0.61 vol pt, Black-Scholes 8.9 vol pt on the stage 3 option set.
- The model-risk numbers come from one date and one product.
- The backtest has 15 trades in two windows. The March 2020 trade is cut at the data end after eight days and is not fully observed.
- End-of-day quotes from one vendor; the index level and option quotes are not exactly synchronous.

Showcase date 2023-01-18 (SPX 3945.1). Monte Carlo results are quoted with one standard error. "Vol pt" denotes 0.01 of implied volatility.

1 Research Design

1.1 Sequence

The project has four stages. Each stage uses only components that the previous stage has already validated, because a pricing error in stage 1 would contaminate every later number.

1. **Numerics.** Do Fourier, finite-difference and simulation methods give the same price for the same contract, and does each converge at its theoretical rate (Fang and Oosterlee, 2008; Andersen, 2008; Longstaff and Schwartz, 2001)?
2. **Calibration.** How well do Heston (Heston, 1993), SABR (Hagan et al., 2002) and Dupire local vol (Dupire, 1994) fit the SPX surface, and which parameters does the surface actually identify?
3. **Model risk.** When several models are fitted to the same surface, how far apart are their prices for path-dependent products, and how much of the difference comes from calibration and how much from dynamics?
4. **Hedging.** Where does the P&L of a delta-hedged option come from, and how much of it can the Greeks explain on real data?

Each stage has a check that does not depend on the market: a closed form, a published reference value, a limiting case, or a synthetic surface whose true parameters are known. Real data enter only after a method has passed on synthetic inputs, so a fit error on SPX can be read as a property of the model rather than of the code.

1.2 Data and cleaning

End-of-day SPX option quotes (16:00 ET) come from optionsDX for January–March 2020 (the COVID crash) and all of 2023 (a calm, rising market). Each row holds the call and put bid and ask at one strike and expiry, with the index level. One loader maps optionsDX, OptionMetrics and Yahoo snapshots to a common schema. Filters, per quote date and expiry:

- two-sided quotes (positive bid, ask above bid, mid at least 0.05), relative spread at most 50%, at least 7 days to expiry;
- standardised log-moneyness $\ln(K/F)/(0.2\sqrt{T})$ between -4 and 2.5, where SPX quotes are liquid;
- out-of-the-money options only: puts below the forward, calls above it.

The forward and discount factor are implied, not assumed. For strikes quoted on both sides, put–call parity gives

$$C(K) - P(K) = D\,(F - K), \tag{1}$$

which is linear in K; a regression on near-the-money strikes gives slope $-D$ and intercept DF. No stale index level, assumed dividend yield or external rate curve enters the surface.

Table 2: Data and audit summary

Item	Value
Raw option quotes (calls and puts)	4,930,870
Quote dates / distinct expiries	311 / 378
Two-sided share of raw quotes	94.7%
Duplicate date–expiry–strike–type rows	0
2023-01-18: raw quotes → cleaned OTM surface	15,106 → 5,498 (44 expiries)
Stage 2 calibration set, 2023-01-18 (8 expiries, 1 month to 1.9 years, up to 25 strikes each)	197 options
Stage 3 calibration set, 2023-01-18 (7 expiries, 1 month to 2.9 years)	170 options
Weekly calibrations, one per calendar week (2020 Q1 / 2023)	14 / 52

1.3 Models

Under the risk-neutral measure with rate r and dividend yield q, the Heston model is

$$\frac{dS_t}{S_t} = (r-q)\,dt + \sqrt{v_t}\,dW_t^S, \qquad dv_t = \kappa(\theta - v_t)\,dt + \xi\sqrt{v_t}\,dW_t^v, \qquad d\langle W^S, W^v\rangle_t = \rho\,dt. \tag{2}$$

Its characteristic function is used in the "little Heston trap" form (Albrecher et al., 2007), which avoids the branch-cut discontinuity of the complex logarithm at long maturities. The COS method (Fang and Oosterlee, 2008) expands the density of $x = \ln(S_T/K)$ in a cosine series on a truncation interval $[a, b]$:

$$V_0 \approx e^{-rT} \sum_{k=0}^{N-1}{}' \operatorname{Re}\left[\varphi\left(\frac{k\pi}{b-a}\right) e^{-ik\pi\frac{a}{b-a}}\right] V_k, \tag{3}$$

where φ is the characteristic function, V_k are the cosine coefficients of the payoff and the prime halves the first term. The interval comes from the cumulants of x, computed numerically from the cumulant generating function.

Local volatility is computed from total implied variance $w(y,T) = \sigma_{\text{imp}}^2 T$ at log-moneyness $y = \ln(K/F)$ (Gatheral, 2006):

$$\sigma_{\text{loc}}^2(y,T) = \frac{\partial_T w}{1 - \frac{y}{w}\partial_y w + \frac{1}{4}\left(-\frac{1}{4} - \frac{1}{w} + \frac{y^2}{w^2}\right)(\partial_y w)^2 + \frac{1}{2}\partial_{yy} w}. \tag{4}$$

Each expiry is fitted with an SVI slice (Gatheral and Jacquier, 2014), so the y-derivatives are analytic; the T-derivative uses linear interpolation in total variance between slices.

2 Three Engines, One Price

2.1 Cross-validation

Table 3 prices three test contracts with every applicable engine. They are different contracts, chosen because each has a trusted reference:

- **Black-Scholes call:** $S = 100$, $K = 95$, $T = 0.75$, $r = 4\%$, $q = 1.5\%$, $\sigma = 25\%$.
- **Heston call:** the Fang and Oosterlee (2008) test case, $S = K = 100$, $T = 1$, $r = q = 0$, with $v_0 = 0.0175$, $\kappa = 1.5768$, $\theta = 0.0398$, $\xi = 0.5751$, $\rho = -0.5711$. The Feller condition is violated, so the variance process reaches zero: the hard case for simulation, handled by the Andersen quadratic-exponential scheme (Andersen, 2008). The published reference value is 5.785155450.
- **American put:** the Longstaff and Schwartz (2001) test case, $S = 36$, $K = 40$, $T = 1$, $r = 6\%$, $\sigma = 20\%$.

Table 3: Cross-validation of pricing engines (Monte Carlo $\pm$ one standard error)

Contract	Reference	COS	Crank–Nicolson	Monte Carlo
BS call ($K = 95$, $T = 0.75$)	12.0056 (closed form)	12.0056	12.0057	12.0042 $\pm$ 0.0058
Heston call ($K = 100$, $T = 1$, $r = q = 0$)	5.7852 (published)	5.7852	–	5.7838 $\pm$ 0.0071
American put ($S = 36$, $K = 40$)	4.4867 (binomial tree)	–	4.4867	4.4758 $\pm$ 0.0065

Agreement is judged against each method's own error: the PDE value differs from the closed form by its discretisation error (10^{-4}), the Monte Carlo values lie within 0.3 standard errors of the reference for the two European contracts, and COS matches the published Heston value to 2×10^{-8} and the Lewis (2001) integral to 10^{-9}. All rows satisfy the no-arbitrage bounds of their own contracts.

For the American put, Longstaff–Schwartz uses a Laguerre basis on 50 exercise dates, with the exercise rule fitted on one set of paths and applied to an independent set. An out-of-sample rule is sub-optimal by construction, so its value is a lower bound for the 50-date Bermudan, which is itself worth less than the American. The Monte Carlo value lies below the PDE value, as it must.

> **Published benchmarks can be wrong.** The finite-difference value 4.478 quoted for this put in Longstaff and Schwartz (2001) is 0.009 below the value from both a 20,000-step binomial tree and the Crank–Nicolson solver (4.4867). Without a second engine the discrepancy would have looked like a bug in the PDE.

2.2 Convergence

Crank–Nicolson is second order in space and time only if the payoff kink does not pollute the solution. The payoff is averaged over the grid cell containing the strike, and the first time steps are fully implicit half-steps (Rannacher start-up; Rannacher, 1984). The Crank–Nicolson amplification factor tends to -1 for stiff modes, so the high-frequency error created by the kink would otherwise never be damped.

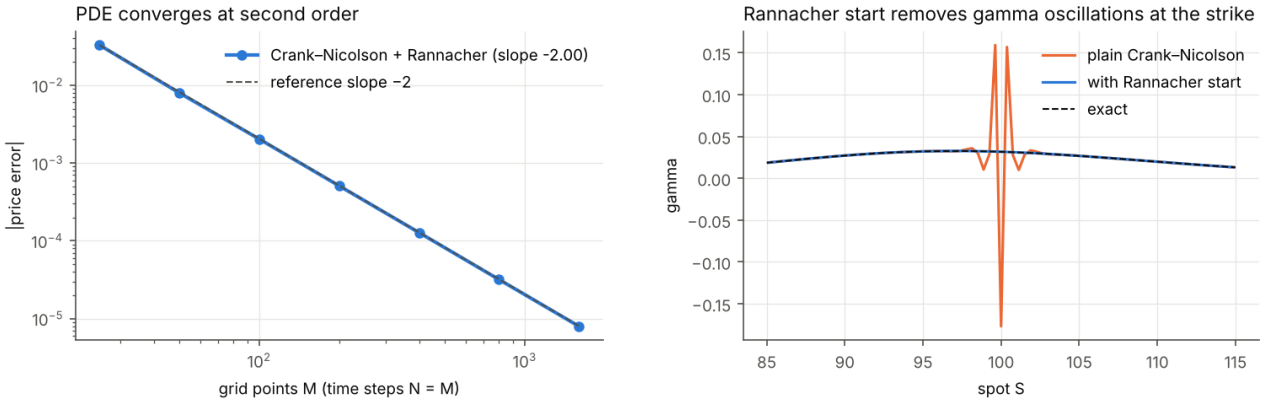


Figure 1: Left: Crank–Nicolson price error against grid size, slope -2.00. Right: with 25 time steps, plain Crank–Nicolson leaves a gamma spike of about 0.2 at the strike; four implicit half-steps remove it (error 10^{-5}).

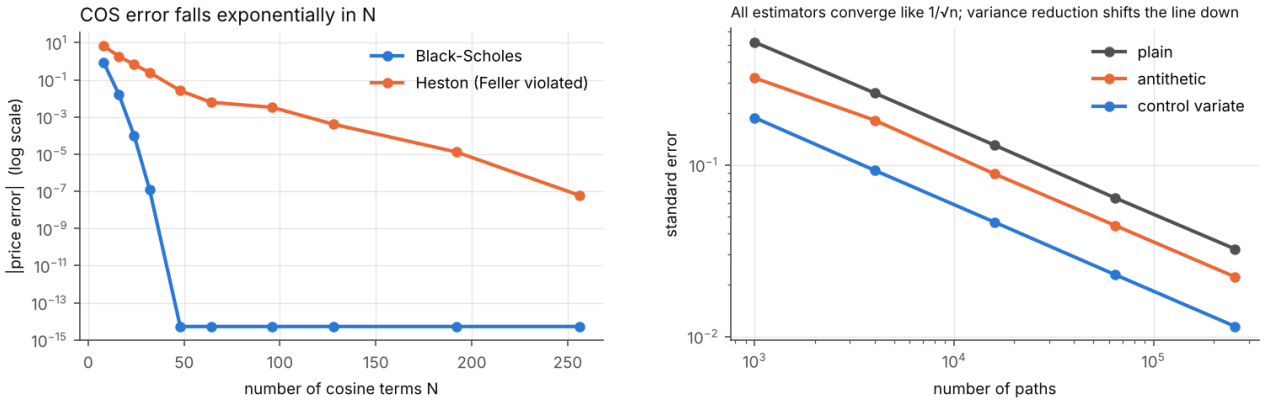


Figure 2: Left: COS converges exponentially in the number of terms; Black-Scholes reaches machine precision by $N \approx 64$, Heston with the Feller condition violated needs more terms. Right: all Monte Carlo estimators converge as $n^{-1/2}$; antithetic sampling cuts the variance 2.1× and the control variate $e^{-rT}S_T$ cuts it 8.0×.

Two further results come from the exotics. A knock-out barrier that falls between grid nodes reduces the PDE to first

order; aligning the grid so that the barrier is a node restores second order. For the autocallable the value jumps at the autocall level and at the knock-in barrier; giving each discontinuity node the average of its two one-sided values restores fast convergence, and the PDE matches Monte Carlo within one standard error.

3 Calibration to the SPX Surface

3.1 Objective and synthetic check

The calibration minimises vega-weighted price errors,

$$\min_{\Theta} \sum_{i} \left(\frac{C_i^{\text{model}}(\Theta) - C_i^{\text{mkt}}}{\max(\mathcal{V}_i,\ \underline{\mathcal{V}})} \right)^2, \tag{5}$$

where $\mathcal{V}_i$ is the Black-Scholes vega and the floor $\underline{\mathcal{V}}$ is 1% of the median vega. To first order each term is an implied-vol error, so the objective is in vol points without a root-finding step inside the optimiser. A bounded trust-region least-squares solver runs from several starting points. On synthetic Heston surfaces the calibrator recovers the true parameters to 10^{-4}; with 0.2 vol points of quote noise added, the fit stops at the noise level (RMSE 0.16 vol pt).

3.2 One date in detail: 2023-01-18

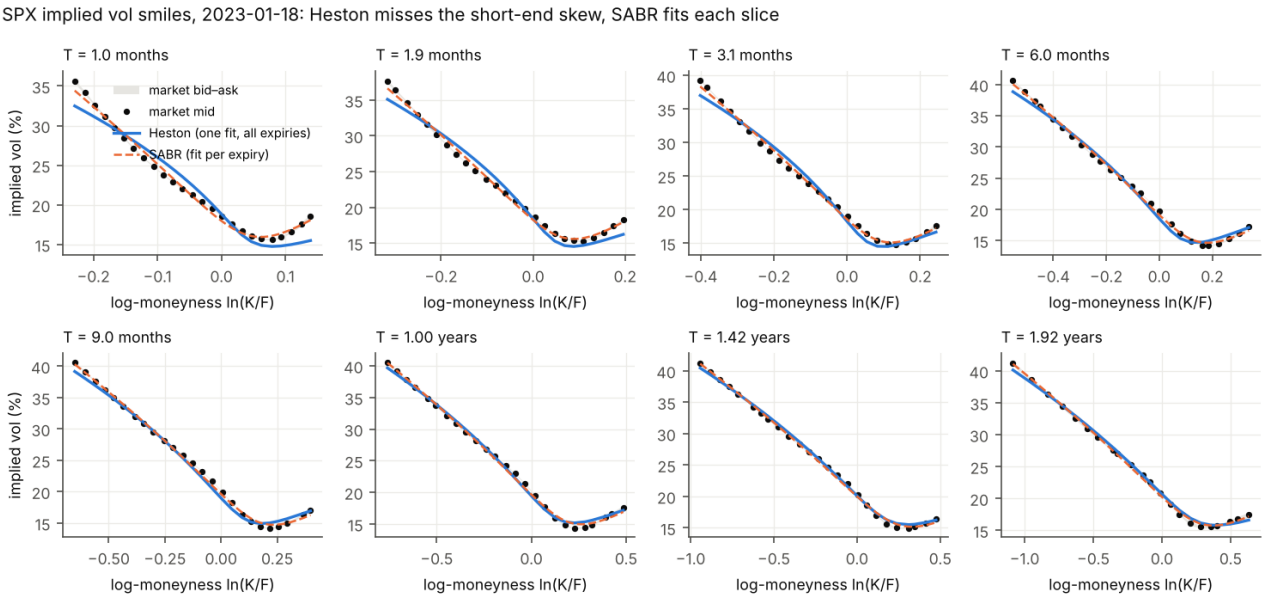


Figure 3: SPX implied vol smiles on 2023-01-18 with the market bid–ask band. One Heston parameter set for all expiries misses the short-dated skew; SABR, refitted on every expiry, follows each slice.

Table 4: Fit quality on 2023-01-18, stage 2 calibration set (implied-vol RMSE against mid)

Expiries	Heston (5 parameters, whole surface)	SABR (3 per expiry, $\beta = 1$)
All	0.89 vol pt	0.43 vol pt
1 month	1.48 vol pt	0.54 vol pt
2 years	0.49 vol pt	0.35 vol pt

The fitted Heston parameters are $v_0 = 0.039$, $\kappa = 1.89$, $\theta = 0.072$, $\xi = 0.97$, $\rho = -0.71$ (Feller ratio $2\kappa\theta/\xi^2 = 0.29$). Equity index surfaces typically violate the Feller condition: the steep skew needs a vol of vol that the condition does not allow.

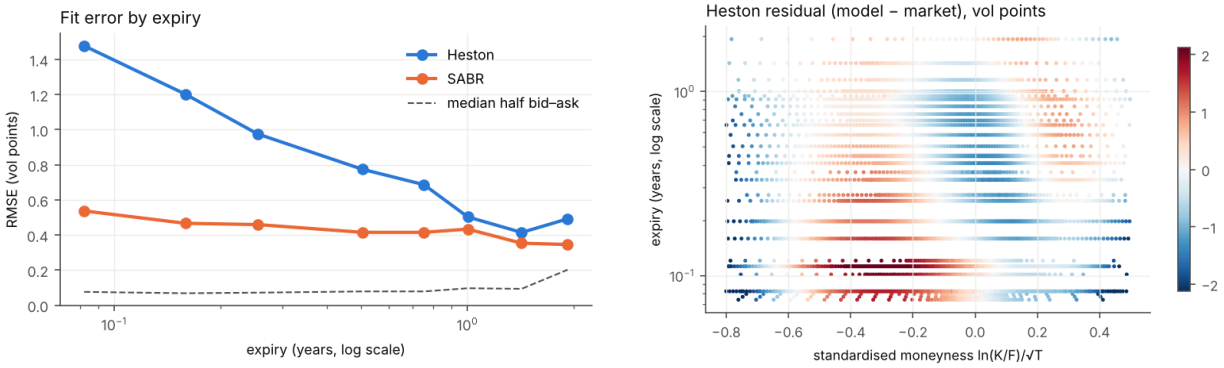


Figure 4: Left: Heston's error falls with maturity; SABR's is roughly flat; both stay above the median half spread. Right: Heston residuals by moneyness and maturity. At short maturities the model is too low in the deep put wing and too high around the money.

> **A model limitation, most likely not an optimiser failure.** In Heston the short-dated skew comes only from the correlated variance diffusion, which needs time to act, so the model struggles to produce the steep one-month SPX skew without an extreme vol of vol. All four optimiser starts reach the same minimum on this date, and in 2023 only 3 of 52 weekly fits had a start ending more than 1% worse. That makes an optimiser failure unlikely, though it does not exclude every effect of the parameter bounds. The conclusion is about Heston: local vol, also a diffusion, fits the one-month slice to 0.43 vol pt. The pattern is the usual motivation for jumps (Bates, 1996) or rough volatility.

SABR fits better for a different reason. Its vol of vol ν falls from 2.25 at one month to 0.58 at two years, roughly as $T^{-0.45}$: the term structure is absorbed into parameters that change with maturity, so SABR interpolates each slice rather than describing the dynamics of the whole surface.

3.3 Local volatility

Dupire local vol built from the SVI slices reproduces a synthetic surface to 0.03 vol pt. On the real surface it reprices the stage 3 option set to 0.61 vol pt (Section 5); most of that is the SVI fit itself, 0.3–0.4 vol pt per slice, with the rest from interpolation between slices and from using constant r and q. Its at-the-money skew is 1.95 times the implied skew, close to the short-maturity rule that implied vol at strike K is roughly the average of local vol between spot and K.

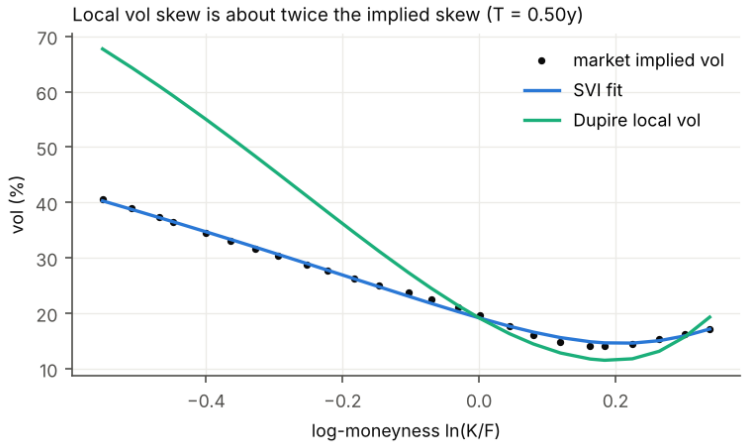


Figure 5: Market implied vol, SVI fit and Dupire local vol at $T = 0.5$ years on 2023-01-18.

4 Parameter Identifiability and Stability

4.1 What a single surface identifies

A good fit says nothing about whether each parameter is pinned down. The direct test is a profile: hold κ fixed on a grid, refit the other four parameters, and record the fit error. It was run first on synthetic surfaces, where the answer is known, and then on the real SPX surface of 2023-01-18.

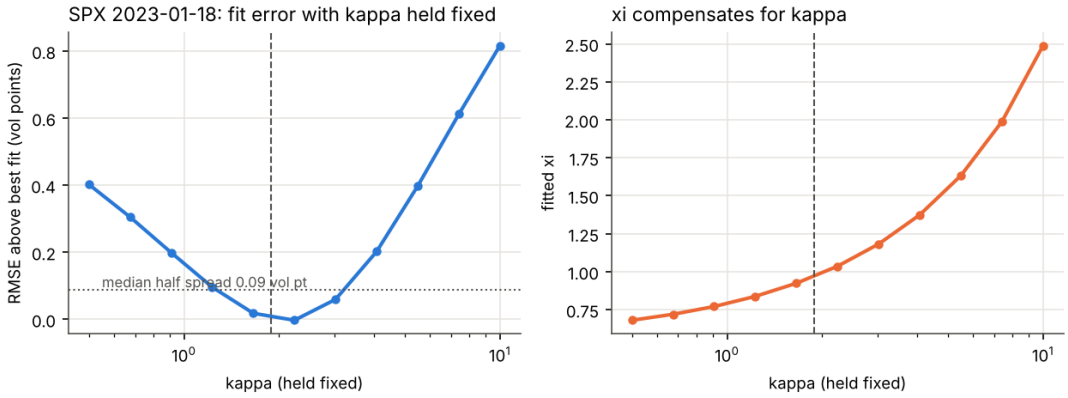


Figure 6: Real SPX surface, 2023-01-18. Left: increase in fit RMSE when κ is held fixed and the other parameters are refitted; the dotted line is the median half bid–ask spread. Right: the refitted ξ rises with κ. The dashed line marks the unconstrained optimum ($\kappa = 1.89$).

Holding κ anywhere between 1.2 and 3.0 raises the RMSE by at most 0.1 vol pt, about one median half spread, while the refitted ξ moves from 0.84 to 1.18. Outside that range the error rises clearly: by 0.4 vol pt at $\kappa = 0.5$ and 0.8 vol pt at $\kappa = 10$. So κ is *weakly* identified, to roughly a factor of 2.5, with ξ compensating along the valley. The synthetic profile (Appendix A) shows the same shape, flatter when only short expiries are used.

4.2 Weekly calibrations

One calibration per calendar week (the Wednesday, or the nearest trading day): 14 dates in 2020 Q1 and 52 in 2023.

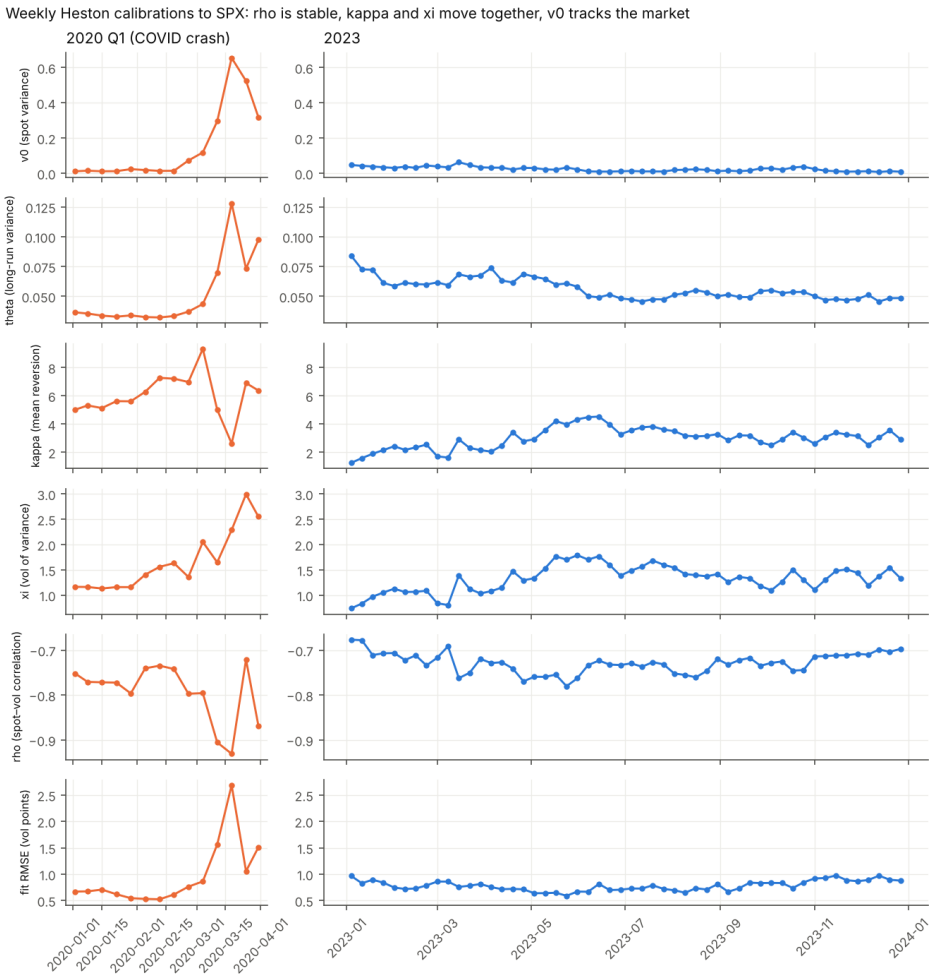


Figure 7: Weekly Heston calibrations to SPX. ρ is stable, κ and ξ move together, and v_0 tracks short-dated implied vol. During the COVID crash every parameter and the fit error move sharply.

Table 5: Heston parameter stability, 52 weekly fits in 2023

Parameter	Mean	Coefficient of variation	Corr. of weekly changes with $\Delta\kappa$
ρ	-0.73	3%	-0.23
θ	0.056	16%	-0.26
ξ	1.32	20%	0.96
κ	2.99	25%	1
v_0	0.024	53%	0.06
Fit RMSE	0.78 vol pt	12%	

- ρ **is stable** (CV 3%): the SPX skew is a persistent feature.
- κ **and** ξ **move together.** Their week-on-week changes correlate at 0.96. On its own this would not prove weak identification, because both could move with the market; together with the single-date profile it indicates that much of the weekly variation in κ is movement along the valley rather than information.
- v_0 **behaves as a state variable**: $\sqrt{v_0}$ has correlation 0.99 with one-month at-the-money implied vol.
- **The crisis degrades the fit.** From 2020-02-19 to 2020-03-18, $\sqrt{v_0}$ rises from 12% to 81%, ρ falls from -0.74 to

-0.93, and the RMSE rises from 0.6 to 2.7 vol pt.

> **Practical consequence.** Risk measures that depend on κ separately from ξ, such as sensitivity to the speed of mean reversion, are poorly determined, and daily recalibration feeds that noise into P&L. Common remedies are to fix κ and recalibrate the rest, or to penalise changes from the previous day's parameters.

5 Exotics and Model Risk

5.1 Products and checks

Down-and-in barrier put. The continuously monitored price has a closed form (Reiner and Rubinstein, 1991). Daily monitoring is handled by the Broadie et al. (1997) correction, which shifts the barrier away from spot by $e^{-\beta\sigma\sqrt{\Delta t}}$ with $\beta = -\zeta(1/2)/\sqrt{2\pi} \approx 0.5826$. For the 3-year 100/60 put the corrected closed form gives 4.92% of spot; daily-monitored Monte Carlo gives 4.93 ± 0.04.

Autocallable. A 3-year note on SPX, notional 100, quarterly observations. If the index is at or above its initial level on an observation date, the note redeems at par plus an 8% p.a. accrued coupon. If it is never called and the index has ever closed below 60% (daily knock-in), the holder receives $100\, S_T/S_0$ when $S_T < S_0$, otherwise par. It is priced by Monte Carlo and by two coupled PDEs (knocked in and not knocked in) stepped once per business day: under Black-Scholes the PDE gives 97.18 against 97.20 ± 0.04 by Monte Carlo, and under local vol both give 96.63. Because the present value is affine in the coupon, $PV(c) = A + cB$, the fair coupon is $c^* = (100 - A)/B$.

5.2 Calibration for a 3-year product

The stage 2 calibration stops at 1.9 years, so stage 3 recalibrates on 7 expiries from one month to 2.92 years (170 options), the longest listed expiry on that date. This gives Heston parameters $v_0 = 0.039$, $\kappa = 1.75$, $\theta = 0.071$, $\xi = 0.96$, $\rho = -0.71$, close to the stage 2 fit. A single $r = 3.93\%$ and $q = 1.49\%$ are fitted to the implied discount and forward curves. Beyond 2.92 years local vol holds each strike's implied vol constant, while Heston extrapolates through its dynamics; the note's last month therefore depends on extrapolation. Black-Scholes uses the at-the-money implied vol at 2.92 years, 20.65%.

5.3 Do the models price the same vanillas?

Fitting to the same surface does not mean fitting it equally well, so every option in the calibration set was repriced under each model, local vol by Crank–Nicolson.

Table 6: Vanilla repricing on 2023-01-18: implied-vol RMSE against mid (vol points)

Expiry (years)	0.08	0.25	0.50	1.00	1.42	1.92	2.92
Black-Scholes	6.52	8.56	9.48	9.67	10.29	9.00	8.31
Heston	1.46	0.98	0.83	0.56	0.40	0.40	0.58
Local vol	0.43	0.41	0.44	0.76	0.41	0.95	0.68
Median half spread	0.08	0.07	0.08	0.10	0.09	0.20	0.37

Overall RMSEs are 8.91 vol pt (Black-Scholes), 0.83 vol pt (Heston) and 0.61 vol pt (local vol). Black-Scholes matches one point of the surface and nothing else, so its coupon difference is a calibration failure, not a statement about dynamics. Heston and local vol both fit to within about a vol point, but neither is inside the spread.

5.4 Separating calibration from dynamics

To split the Heston–local vol gap, a third model is built: local vol constructed from *Heston's* implied vols on the same expiries and strikes. It reprices Heston's vanillas to 0.07 vol pt, so it differs from Heston only in dynamics, and from market local vol only in the vanillas it was built from.

Table 7: Autocallable under three models fitted to the same surface, and the calibration–dynamics split

Model	Fair coupon	P(call, 1st date)	P(KI loss)	Life (y)	DIP (% S_0)
Black-Scholes	14.03%	50.3%	8.2%	0.96	4.93
Heston	15.17%	63.8%	8.3%	0.73	9.32
Local vol, market smiles	15.95%	58.8%	8.9%	0.73	8.43
Local vol, Heston smiles	15.94%	63.6%	–	–	–
Dynamics effect (LV on Heston smiles − Heston): $+0.77$ coupon pts					
Calibration effect (LV on market smiles − LV on Heston smiles): $+0.01$ coupon pts					

Fair-coupon Monte Carlo standard errors are 0.10–0.14 points (100,000 paths), so each effect carries an error of about 0.19 points. DIP: 3-year down-and-in put, strike 100%, barrier 60%, daily monitoring.

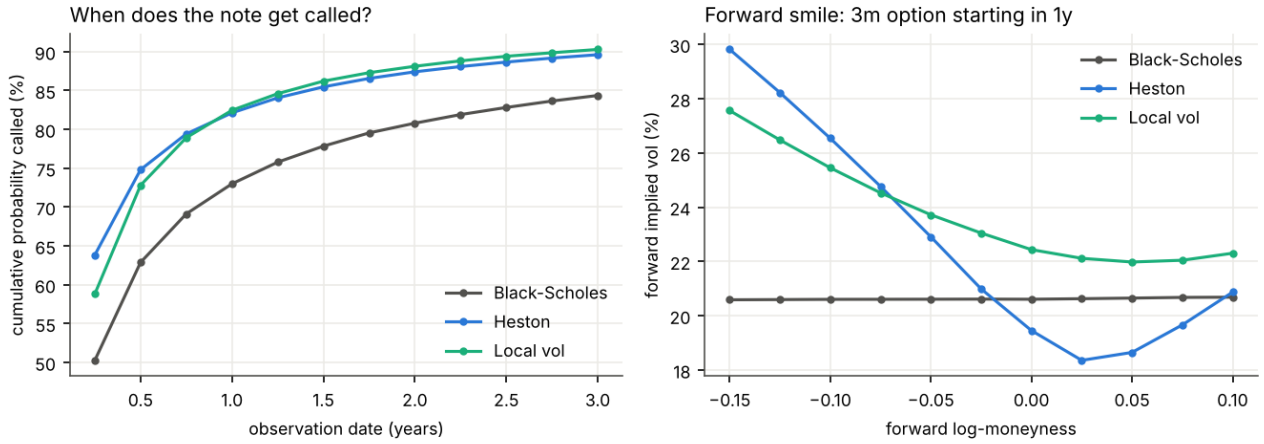


Figure 8: Left: cumulative probability that the note has been called by each observation date. Right: forward smile, the implied vol of a 3-month option starting in one year, computed from each model's paths. Local vol's forward skew is flatter than Heston's.

> **Interpretation.** On this date and for this note, almost all of the 0.78-point Heston–local vol gap survives when the two models are given the same vanillas, so it comes from dynamics rather than from their different calibration errors. That is consistent with the forward-smile difference in the figure: local vol's skew flattens as time passes, Heston's persists, and the autocall and knock-in features depend on those future smiles. The Black-Scholes gap is of a different kind: with no skew it prices the 60% knock-in put at about half the level of the skew-consistent models.

The result rests on one surface and one product, and the two calibration errors happen to have similar coupon impact here; a different date or payoff could change the split.

6 Greeks and the Sources of Hedging P&L

6.1 Greeks by simulation

Monte Carlo Greeks were implemented three ways. The pathwise estimator differentiates along each path, $\Delta = \mathbb{E}[e^{-rT}\mathbf{1}\{S_T > K\}S_T/S_0]$, and fails for a digital option, whose payoff has zero derivative almost everywhere. The likelihood-ratio estimator moves the derivative onto the density, $\Delta = \mathbb{E}[e^{-rT} f(S_T)\, Z/(S_0\sigma\sqrt{T})]$, and works for any payoff at the cost of higher variance. A mixed estimator, likelihood ratio applied to the pathwise delta, gives gamma. All three match the closed forms within three standard errors.

6.2 The hedged P&L identity

For a short option that is delta-hedged and self-financed, the Black-Scholes PDE turns theta and financing into $\frac{1}{2}\Gamma S^2\sigma_{\text{imp}}^2\,dt$, so to second order one day's P&L is

$$d\Pi \approx \underbrace{\tfrac{1}{2}\Gamma S^2\left(\sigma_{\text{imp}}^2\,dt - \left(\tfrac{dS}{S}\right)^2\right)}_{\text{gamma/theta}} - \underbrace{\mathcal{V}\,d\sigma_{\text{imp}}}_{\text{vega}} - \underbrace{\left(\text{vanna}\cdot dS\,d\sigma_{\text{imp}} + \tfrac{1}{2}\text{volga}\cdot d\sigma_{\text{imp}}^2\right)}_{\text{vanna/volga}} + \varepsilon. \tag{6}$$

A short-gamma position earns when realised variance is below implied and loses when it is above.

6.3 Simulation: the cost of discrete hedging

In a Black-Scholes world with a 3-month at-the-money call and $\sigma = 20\%$, the hedging error from rebalancing N times should follow Kamal and Derman (1999):

$$\text{sd}(\text{P\&L}) \approx \sqrt{\frac{\pi}{4}}\,\frac{\mathcal{V}\,\sigma}{\sqrt{N}}. \tag{7}$$

With 63 daily rebalances the simulated standard deviation is 0.434, against 0.441 from the formula (per 100 notional). With 5 bp transaction costs, cost grows like $\sqrt{N}$ while hedging error falls like $1/\sqrt{N}$, giving an optimal frequency near 252 rebalances here.

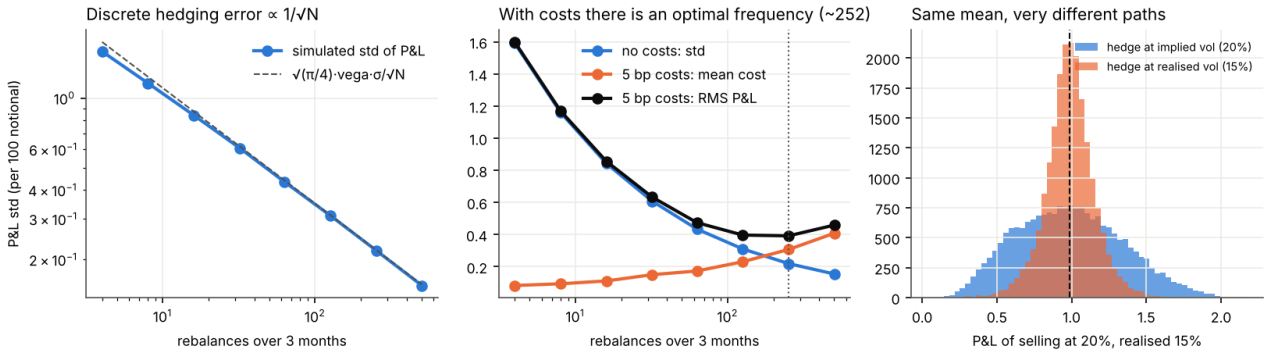


Figure 9: Left: discrete-hedging error scales as $N^{-1/2}$ and matches Kamal–Derman. Centre: with 5 bp costs the RMS P&L has a minimum. Right: selling at 20% implied when realised is 15%; hedging with implied-vol and realised-vol deltas gives the same mean (0.99) but different dispersion (0.35 vs 0.16).

6.4 Real SPX backtest

On the first trading day of each month one 1-month at-the-money SPX call is sold at mid (strike nearest the implied forward, expiry nearest 30 days), hedged daily in the index, re-marked daily at mid, and closed on the last quote date before expiry. Each day's P&L is split two ways.

- **Taylor attribution:** the identity above, with Greeks at the start of the day.

- **Full revaluation:** the option is repriced in three steps. (a) Time passes and the index moves at fixed implied vol, with the forward rolling at constant carry; together with the hedge this is the realised-versus-implied variance term. (b) The forward and discount factor move to their own day-end values: the forward–index basis. (c) Implied vol moves to its day-end value. The three pieces add to the total exactly.

Table 8: SPX hedging backtest, index points per call, no transaction costs

		2020 Q1 (3 trades)	2023 (12 trades)
Hedged P&L		-127.8	$+31.8$
Full revaluation	spot and time at fixed vol	-84.7	$+74.2$
	implied-vol re-marks	-41.3	-34.2
	forward–index basis	-1.8	-8.2
Taylor	gamma/theta	-96.4	$+73.6$
	vega	-65.5	-75.6
	vanna/volga	$+10.6$	$+2.1$
	residual	$+23.5$	$+31.7$
Mean entry implied vol / realised vol		17.2% / 36.3%	15.6% / 12.1%
Daily P&L variance explained by Taylor terms		97%	84%

Daily fit and cumulative attribution are different questions. The Taylor terms explain 97% of daily P&L variance in 2020 Q1 and 84% in 2023 (91% pooled, a figure dominated by the large 2020 days). Yet in 2023 the Taylor components sum to $+0.1$ against a total of $+31.8$: almost the entire cumulative profit sits in the residual. Small errors with a consistent sign accumulate even when the daily fit is good. The full revaluation removes the ambiguity.

Where the Taylor residual comes from. Comparing each Taylor term with its full-revaluation counterpart locates the 2023 residual of $+31.7$ points:

- **Vol re-marks, $+39.4$.** Full revaluation gives -34.2 where vega plus vanna/volga give -73.5. Of this, $+25.0$ comes from the last three days before expiry. There, the fixed strike's implied vol changes by 5–25 vol points per day, and vega itself changes sharply with both vol and time, so a second-order expansion in $d\sigma$ breaks down. Implied vols of 1–3 day options are also sensitive to the time convention near an AM-settled expiry.
- **Forward–index basis, -8.2.** The ratio of the parity-implied forward to the reported index level moves by about 6 bp per day (standard deviation) against roughly 1 bp of carry. This is mostly asynchrony between the index print and the option quotes. The term accounts for 15% of daily P&L variance in 2023 and is the main reason the daily fit is lower than in 2020.
- **Spot and time, $+0.6$.** The gamma/theta Taylor term matches full revaluation closely in a calm year. In 2020 Q1 the gap is $+11.7$, from third-order spot moves on large days.

Reading the backtest. In 2023 implied vol averaged 15.6% against 12.1% realised, and the variance term earned $+74.2$ points over 12 trades; implied-vol re-marks gave back -34.2. The re-mark loss is consistent with a fixed strike sliding down the skew in a rising market, so that its implied vol rises even when the smile does not move. In 2020 Q1 realised vol of 36% against 17% implied cost -84.7 in the variance term and -41.3 in re-marks over three trades.

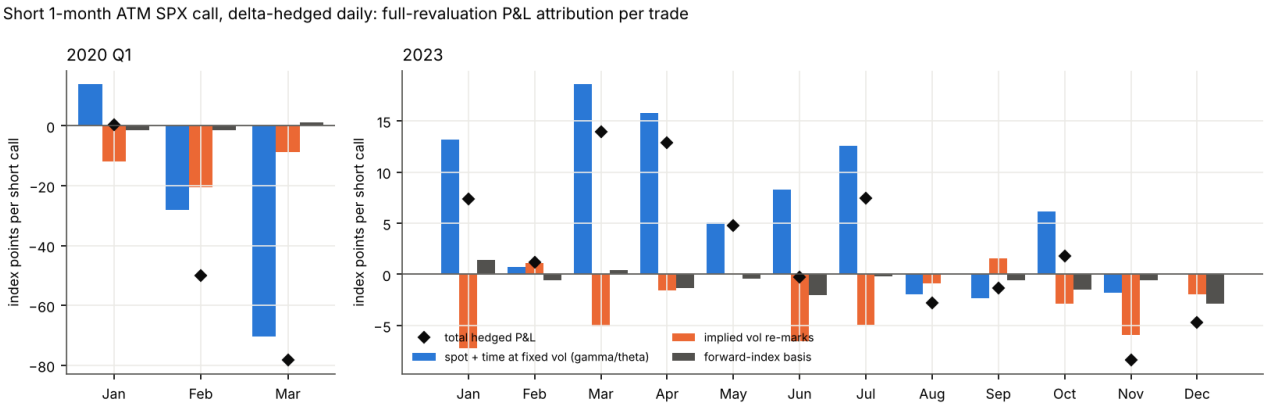


Figure 10: Full-revaluation attribution per monthly trade. In 2023 the realised-versus-implied variance term (blue) is mostly positive and implied-vol re-marks (orange) give back part of it.

Inference limits. Fifteen trades in two chosen windows cannot estimate the return of selling volatility. The March 2020 trade ends after eight days at the data cut-off, so its outcome is not fully observed; the loss could have grown or partly recovered. The backtest shows *where* hedged P&L came from, not how much a strategy would earn, and it ignores bid–ask costs on the option and the hedge.

7 Evidence Summary

1. **Numerical correctness: strong, within scope.** Three engine types agree on the three test contracts within discretisation or statistical error, match published or tree references, and converge at their theoretical rates.
2. **Heston's short-end misfit is a model limitation: strong for Heston.** Synthetic recovery to 10^{-4}, stable optimiser results and structured residuals point to the model rather than the fitting. It does not extend to other diffusions.
3. **Weak identification of κ: strong on one surface, supported over the year.** The real-surface profile is flat to within one half spread over $\kappa \in [1.2, 3.0]$; weekly κ and ξ changes correlate at 0.96.
4. **Heston–local vol coupon gap from dynamics: supportive.** A controlled experiment attributes $+0.77$ of $+0.78$ points to dynamics on one date and one note.
5. **Hedging attribution: descriptive.** The full-revaluation split is exact; the Taylor residual is located; the economic reading rests on 15 trades.

The contribution is not a new model. It is a library in which each number has an independent check, applied so that the differences that remain, between models and between the hedge and the market, can be traced to a cause.

8 Limitations and Next Steps

8.1 Limitations

- **Data:** end-of-day mids from one vendor; the index level and the option quotes are not synchronous.
- **Rates and dividends:** one r and q per date for local vol and the autocallable. The implied forward curve deviates from this by up to about 50 bp across expiries.
- **Fit quality:** no model reprices the surface inside the spread; local vol inherits the SVI slice errors.

- **Heston PDE:** no 2D solver, so Heston prices are cross-checked by COS, Monte Carlo and a published value only.
- **Scope:** the model-risk split uses one date and one product; the backtest uses 15 trades and no costs.

8.2 Next steps, in priority order

1. Repeat the calibration–dynamics split on the weekly 2023 surfaces, to see whether it holds over time.
2. Hedge with the parity-implied forward (or index futures) instead of the index print, removing the basis noise.
3. Close backtest trades several days before expiry, or attribute the final days separately, and add bid–ask costs.
4. Use term structures of r and q in local vol and in the autocallable.
5. Only then extend the model set: jumps for the short end, stochastic local vol for the autocallable.

A Synthetic Identifiability Check

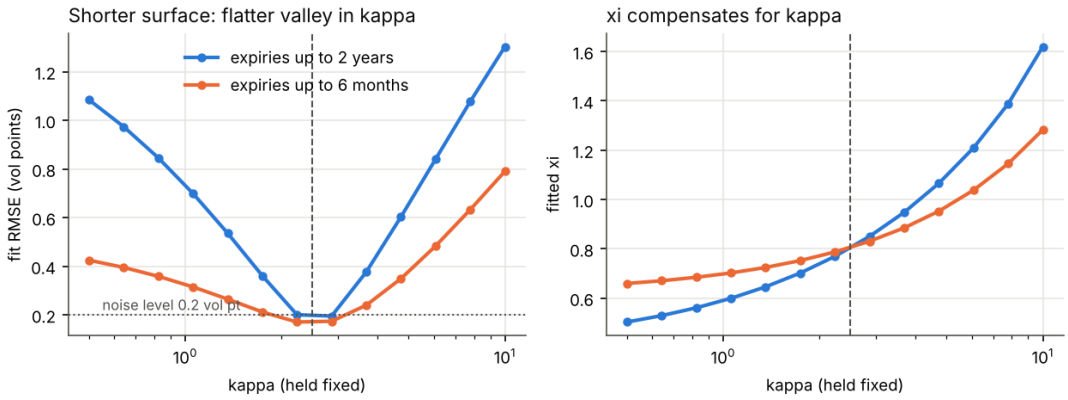


Figure 11: Synthetic Heston surface with 0.2 vol pt quote noise, true $\kappa = 2.5$. Left: fit RMSE with κ held fixed, flatter when only expiries up to six months are used. Right: the refitted ξ rises with κ along the valley.

B Reproducibility

All code, tests, derivations and stage scripts are at github.com/kexinzhang2026/equity-derivatives-pricing. Raw optionsDX files are not redistributed; the README explains how to obtain them.

- 71 unit tests (pytest) run in continuous integration on every push. They use synthetic inputs only and need no market data.
- **Black-Scholes:** textbook values, put–call parity, no-arbitrage bounds, Greeks against finite differences and against the Black-Scholes PDE.
- **Heston COS:** the published Fang–Oosterlee value to 10^{-7}, the Lewis (2001) integral to 10^{-9}, and the Black-Scholes limit as $\xi \to 0$ at rate $O(\xi^2)$.
- **PDE:** convergence order on Europeans; American put against a binomial tree; barrier order with the grid aligned.
- **Monte Carlo:** each estimator within three standard errors of the analytic value; Longstaff–Schwartz below the PDE value.
- **Calibration:** implied forward recovered from synthetic chains; Heston parameter recovery with and without noise; Dupire round trip on a synthetic surface.

- **Exotics and hedging:** barrier closed form against PDE and Monte Carlo; autocallable PDE against Monte Carlo; Kamal–Derman scaling; the full-revaluation pieces sum to the total.
- Every formula in the code is derived in `docs/derivations/01–10`. One script per stage in `scripts/` regenerates its tables and figures (stage 1 cross-validation, stage 2 calibration and κ profile, stage 3 model risk, stage 4 hedging).

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